Connectome-Based Predictive Modelling of Cognitive Performance in Psychiatric Disorders

NTU-SG, Leong Rui Ying Clarisse, U2331671A

A patient walks into your clinic.

- He is diagnosed with schizophrenia
- You prescribe antipsychotics to manage his symptoms
- 6 months later, his psychosis is controlled but he still can't
  - Hold a job
  - Follow instructions
  - Live independently

Why? Because antipsychotics target psychosis, not cognition. Cognitive deficits like impaired working memory and attention persist even when symptoms are managed.

Despite decades of research, no neuroimaging biomarkers have been adopted in routine psychiatric clinical practice (Abi-Dargham et al., 2023).

The Scale of the Problem

The Burden

Mental disorders affect over 1 billion people worldwide, accounting for 16% of global disease burden and USD 5 trillion in economic losses annually (Arias et al., 2022).

The Gap

Diagnosis still relies on symptom-based criteria with no objective biological measures to guide treatment, and we have no way to predict which patients will develop cognitive deficits or how severe they will be.

Project Aim

To determine whether resting-state functional connectivity can predict individual cognitive performance transdiagnostically across schizophrenia, bipolar disorder, ADHD, and healthy controls.

Every brain has a unique connectivity fingerprint.

Resting-state fMRI
Scans the brain at rest, capturing its intrinsic organisation rather than task performance.

Connectome
A map of functional connections between brain regions, unique to each individual.

Why resting-state?
Cognitive deficits in psychiatric disorders are present all the time, not just during a task.

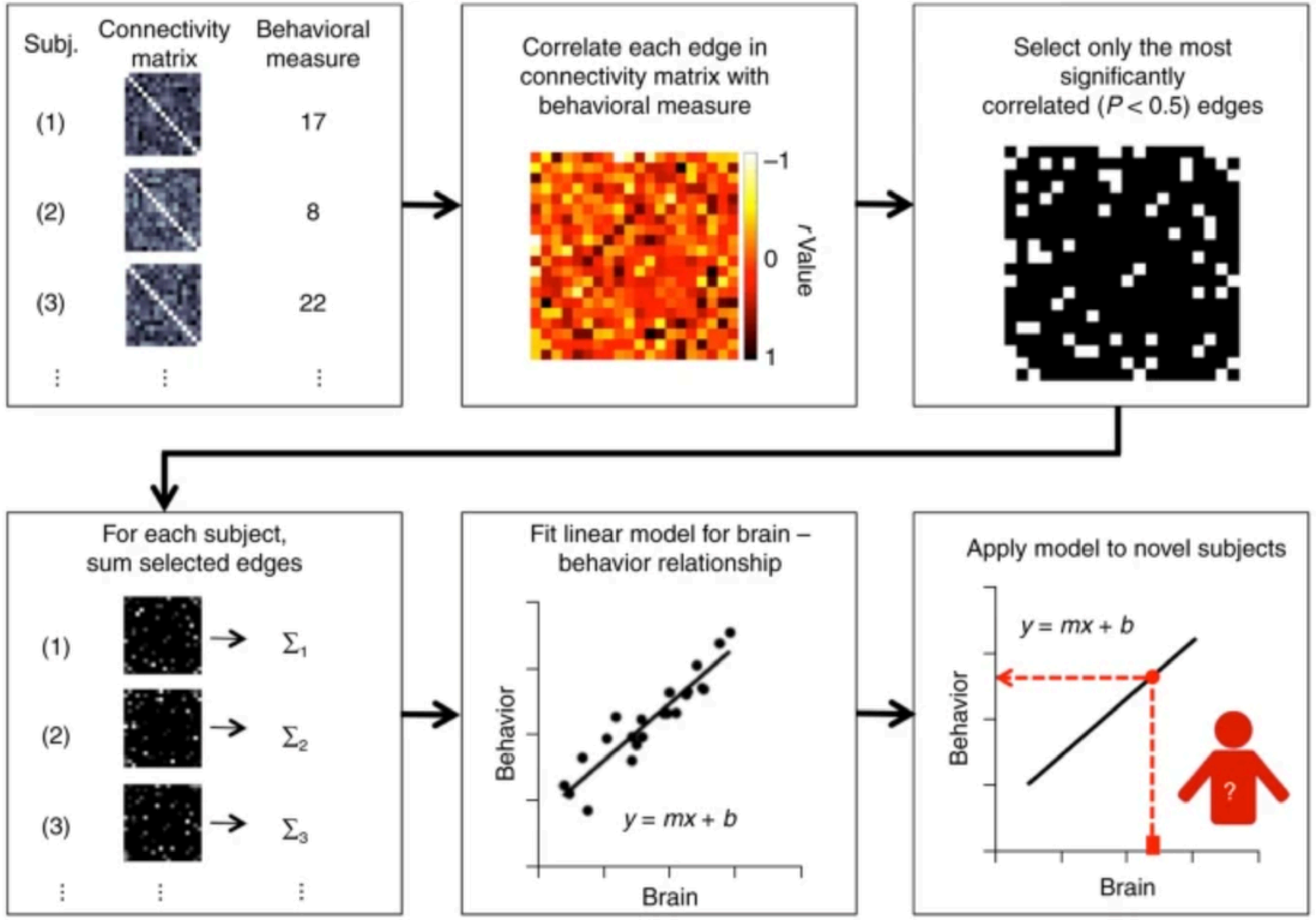
Transdiagnostic
Studying what schizophrenia, bipolar disorder, and ADHD share at the brain level rather than treating them separately.

Can a brain scan predict how well someone thinks?

Connectome-Based Predictive Modelling (CPM)

CPM uses whole-brain connectivity patterns to predict individual cognitive performance through **leave-one-out cross-validation** (Shen et al., 2017.)

Figure 1: Schematic of CPM.



Barron et al. (2021): task-based only. Zhu et al. (2021): one domain, static model nonsignificant. Lv et al. (2025): inhibitory control only.

This project: resting-state CPM across five cognitive domains transdiagnostically

What am I predicting and how?

Dataset:

UCLA Consortium for Neuropsychiatric Phenomics (Poldrack et al., 2016)
Preprocessed resting-state fMRI, fMRIPrep pipeline (Esteban et al., 2019)
N = 247: schizophrenia (n=50), bipolar disorder (n=49), ADHD (n=23), healthy controls (n=125)

Five cognitive domains

- Working Memory
- Attention
- Verbal Memory
- Processing Speed
- Inhibitory Control

Applying the relevant modules!

- Functional connectivity in fMRI
- Python and R for data analysis
- Machine learning basics
- Data visualisation
- Open data and BIDS
- Git/GitHub

If resting-state connectivity predicts cognition across disorders, it changes how we think about clinical assessment.

What this means clinically:

Rather than running a full neuropsychological battery, a single non-invasive resting-state scan could one day tell a clinician which cognitive systems are at risk in a patient regardless of their diagnosis.

Precision psychiatry, where treatment is guided by an individual's brain, not just their diagnosis!

Citations

Abi-Dargham, A., Moeller, S. J., Ali, F., DeLorenzo, C., Domschke, K., Horga, G., Jutla, A., Kotov, R., Paulus, M. P., Rubio, J. M., Sanacora, G., Veenstra-VanderWeele, J., & Krystal, J. H. (2023). Candidate biomarkers in psychiatric disorders: state of the field. World Psychiatry, 22(2), 236–262. https://doi.org/10.1002/wps.21078

Arias, D., Saxena, S., & Verguet, S. (2022). Quantifying the Global Burden of Mental Disorders and Their Economic Value. EClinicalMedicine, 54(101675), 101675. https://doi.org/10.1016/j.eclinm.2022.101675

Barron, D. S., Gao, S., Dadashkarimi, J., Greene, A. S., Spann, M. N., Noble, S., Lake, E. M. R., Krystal, J. H., Constable, R. T., & Scheinost, D. (2020). Transdiagnostic, Connectome-Based Prediction of Memory Constructs Across Psychiatric Disorders. Cerebral Cortex, 31(5), 2523–2533. https://doi.org/10.1093/cercor/bhaa371

Esteban, O., Markiewicz, C. J., Blair, R. W., Moodie, C. A., Isik, A. I., Erramuzpe, A., Kent, J. D., Goncalves, M., DuPre, E., Snyder, M., Oya, H., Ghosh, S. S., Wright, J., Durnez, J., Poldrack, R. A., & Gorgolewski, K. J. (2019). fMRIPrep: a robust preprocessing pipeline for functional MRI. Nature Methods, 16(1), 111–116. https://doi.org/10.1038/s41592-018-0235-4

Finn, E. S., Shen, X., Scheinost, D., Rosenberg, M. D., Huang, J., Chun, M. M., Papademetris, X., & Constable, R. T. (2015). Functional connectome fingerprinting: identifying individuals using patterns of brain connectivity. Nature Neuroscience, 18(11), 1664–1671. https://doi.org/10.1038/nn.4135

Lv, Q., Wang, X., Kang, N., Wang, X., & Lin, P. (2025). Transdiagnostic Connectome-Based Prediction of Response Inhibition. Human Brain Mapping, 46(3). https://doi.org/10.1002/hbm.70158

Poldrack, R. A., Congdon, E., Triplett, W. T., Gorgolewski, K. J., Karlsgodt, K. H., Mumford, J. A., Sabb, F. W., Freimer, N. B., London, E. D., Cannon, T. D., & Bilder, R. M. (2016). A phenome-wide examination of neural and cognitive function. Scientific Data, 3(1). https://doi.org/10.1038/sdata.2016.110

Shen, X., Finn, E. S., Scheinost, D., Rosenberg, M. D., Chun, M. M., Papademetris, X., & Constable, R. T. (2017). Using connectome-based predictive modeling to predict individual behavior from brain connectivity. Nature Protocols, 12(3), 506–518. https://doi.org/10.1038/nprot.2016.178

Zhu, J., Li, Y., Fang, Q., Shen, Y., Qian, Y., Cai, H., & Yu, Y. (2021). Dynamic functional connectome predicts individual working memory performance across diagnostic categories. NeuroImage: Clinical, 30, 102593. https://doi.org/10.1016/j.nicl.2021.102593